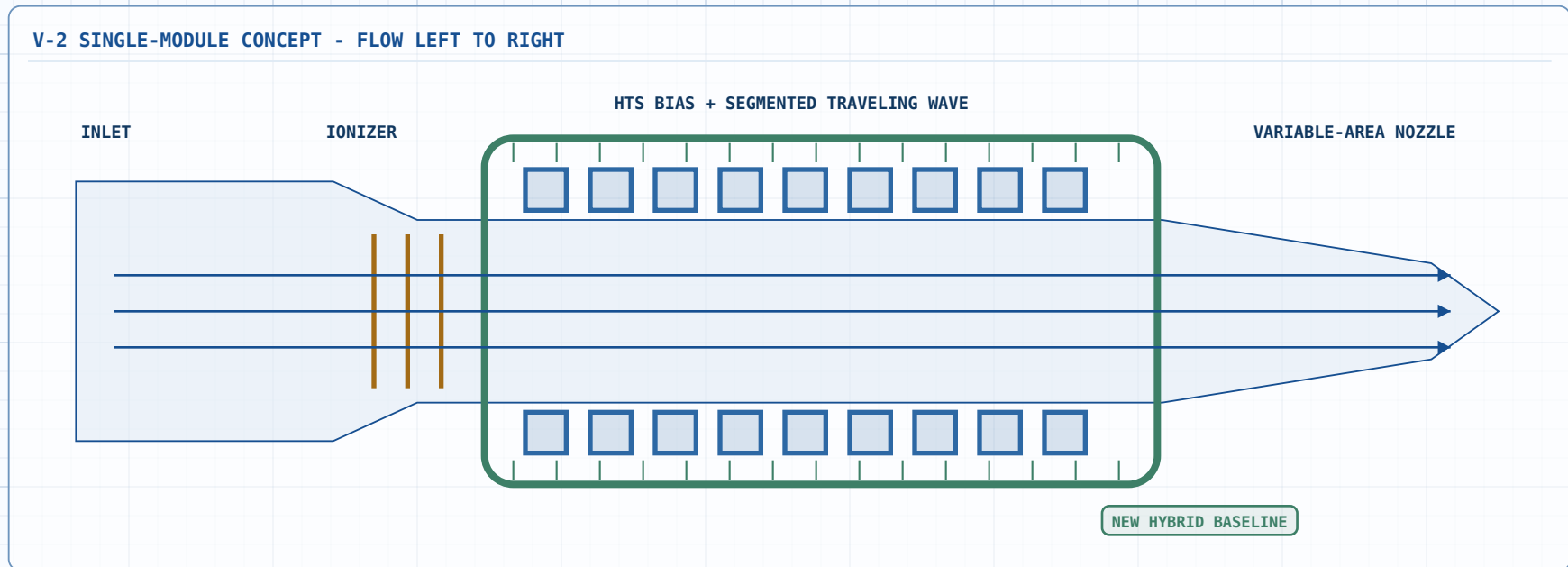


CRTFE V-2

HYBRID MHD PROPULSION BASELINE

HTS-BIASED / TRAVELING-WAVE / EXTERNALLY IONIZED / NOZZLE-COUPLED



CONCEPT / TEST DEFINITION ONLY - NOT A FABRICATION DRAWING - NOT FLIGHT READY

No energized high-voltage, cryogenic, pressure, plasma, combustion, tethered-flight, or crewed test is authorized by this package.

01 CONFIGURATION DEFINITION

V-2 NAMES THE PROPULSION ARCHITECTURE; V5 REMAINS THE VEHICLE REFERENCE

FROZEN V-2 DESCRIPTION

A shielded, closed-loop-controlled, HTS-biased traveling-wave MHD system that externally ionizes atmospheric flow, transfers electromagnetic energy into pressure and momentum, and uses a variable-area nozzle to produce measurable exit momentum.

BASELINE DECISION

VERSION TAXONOMY

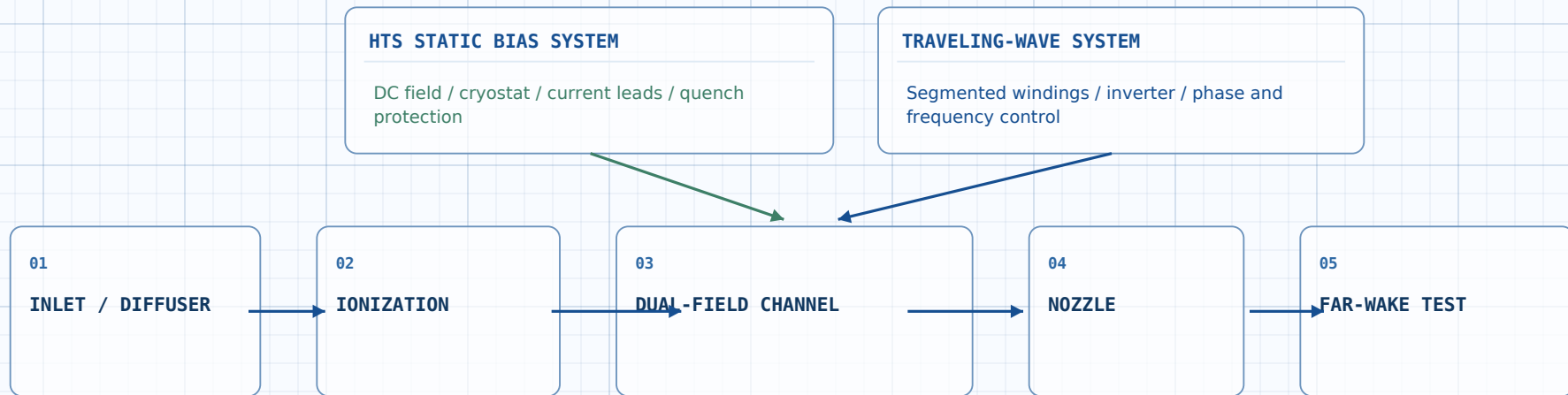
V-2A	0.1 m2 x 0.5 m bench channel	Conductivity + interaction
V-2B	One approx. 0.6 m2 module	Full force/power closure
V-2B+	Two synchronized modules	Phase-lock / interference
V-2C	Four synchronized modules	Vehicle integration
V-2D	Shielded representative assembly	Structural / thermal qualification

STATUS	MEANING	CURRENT V-2 USE	RELEASE RULE
PROJECT VALUE	Carried from supplied CRTFE material	Vehicle/module envelopes	Retain traceability
MODELED PLACEHOLDER	Input used for parametric screening	1.8 T, 150 S/m, 5 kW auxiliary	Replace with measurement/design
PREDICTED	Output of an unvalidated model	Flow, force and power results	Do not claim as performance
MEASURED	Observed with calibrated instrumentation	None yet for V-2	Report uncertainty and conditions
RELEASED	Approved for procurement/fabrication	None	Requires engineering authority

SEPARATE AXES: CLAIM EVIDENCE ABOVE; REQUIREMENT MATURITY = DRAFT -> ALLOCATED -> APPROVED -> IMPLEMENTED -> VERIFIED -> VALIDATED.

02 DUAL-FIELD FUNCTIONAL ARCHITECTURE

THE STATIC BIAS SYSTEM AND TRAVELING-WAVE SYSTEM REMAIN PHYSICALLY AND ELECTRICALLY SEPARABLE



FIELD / FORCE MODEL

$$E_{\text{total}} = E_{\text{induced}} + E_{\text{applied}}$$

$$B_{\text{total}} = B_{\text{HTS}} + B_{\text{wave}}(x, t)$$

$$J = \text{sigma_tensor} [E_{\text{total}} + u \times B_{\text{total}}]$$

$$f_{\text{EM}} = J \times B_{\text{total}}$$

$$q_{\text{field}} = J \text{ dot } E_{\text{total}}$$

$$v_{\text{wave}} = f \text{ lambda}$$

$$\text{slip} = (v_{\text{wave}} - u) / v_{\text{wave}}$$

CONTROL AND SAFETY PARTITION

- Deterministic control calculates, commands and enforces approved setpoints.
- Independent protection hardware owns all trip thresholds.
- THE ARC may submit authenticated, non-executable ground-test intent only; it has no raw actuator authority.
- Hardwired trips never wait for ARC, network services or cryptographic audit.
- Claimed force must survive polarity reversal, field-only, plasma-only, blank-flow and cable-decoupling controls.

03 WHAT TRANSFERS AND WHAT IS NEW

USE PROVEN COMPONENTS; EXPERIMENTALLY VALIDATE THE NEW INTEGRATION

AREA	TRANSFERABLE FOUNDATION	V-2 ENGINEERING WORK	STATUS
HTS magnet	Conduction cooling, HTS tape, current leads, cryogenic sensors and quench methods	Large-bore field volume, support loads, AC exposure, mass and shielding	CANDIDATE
Traveling wave	Polyphase windings, phase control and winding termination methods	Geometry matched to finite duct, tensor conductivity and force direction	NEW DESIGN
Ionization	Pulsed, RF, microwave and specialized electron-beam precedents	Uniform atmospheric conductivity with acceptable net energy cost	UNVALIDATED
MHD interaction	$\mathbf{J} \times \mathbf{B}$ force and $\mathbf{J} \cdot \mathbf{E}$ energy accounting	Coupled transient electromagnetics, compressible flow and nozzle solution	MODEL 0.3
Nozzle	Pressure-to-velocity conversion and duct instrumentation	Area law, losses, wall load, thermal response and far-wake closure	MODELED
Protection	Cryostats, fragment shields, fire barriers and interlocks	Field-transparent layered enclosure without shorted conductive loops	NEW DESIGN
Controls	Deterministic power control, sensing and fault isolation	Real-time coordination of conductivity, slip, field and pressure	NEW DESIGN

NOVELTY IS NOT PROOF: THE COMBINATION MAY BE NEW, BUT PERFORMANCE AND PATENTABILITY REQUIRE TESTS AND PRIOR-ART REVIEW.

04 NUMERICAL BASELINE AND LABELS

ALL VALUES BELOW RETAIN THEIR EVIDENTIARY STATUS

PARAMETER	VALUE	STATUS	USE / LIMIT
V-2A channel	0.1 m2; L = 0.5 m	PROJECT TEST TARGET	Conductivity and interaction bench
V-2B interaction area	Approx. 0.6 m2	PROJECT VALUE	Single-module screening envelope
Interaction length	0.5 m	PROJECT VALUE	Finite-length field region
HTS bias field	1.2-1.8 T	MODELED TARGET	Requires magnet trade and field map
Uniform conductivity	150 S/m	MODELED PLACEHOLDER	Must be replaced with sigma(x,t)
Bench airflow	10-30 m/s	PROJECT TEST TARGET	V-2A non-flight test range
Model 0.2 mass flow	20.25 kg/s	PREDICTED	Based on 0.6 m2 inlet and stated conditions
Model 0.2 exit velocity	55.26 m/s	PREDICTED	With 0.3 m2 exit; not net thrust closure
Model 0.2 field power	39.75 kW	PREDICTED	100% inverter efficiency placeholder
Auxiliary power	5.0 kW	UNVALIDATED	Does not yet include real ionizer/cryo budget

1.8 T FIELD-VOLUME LOWER-BOUND CHECK

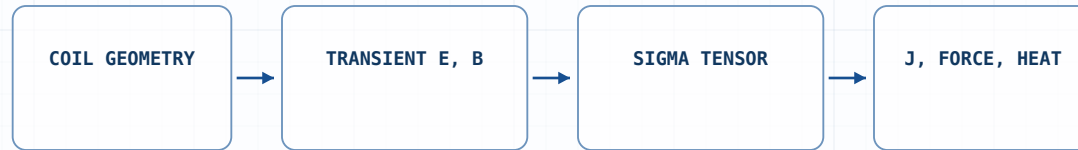
$u_B = B^2 / (2 \mu_0) = 1.29 \text{ MJ/m}^3$ | $p_B = 1.29 \text{ MPa}$ | $V = 0.6 \times 0.5 = 0.30 \text{ m}^3$ | $U_{ideal} = 0.387 \text{ MJ/module}$

Stored energy is not continuous power. Actual coil energy, structure and leakage depend on geometry; four idealized modules imply about 1.55 MJ.

05 FINITE-COIL ELECTROMAGNETIC MODEL

G2 IS A STAGED VERIFICATION AND VALIDATION GATE - NOT ONE FIELD-MAP CHECK

REQUIRED MODEL CHAIN



Outputs: $B(x,y,z,t)$, $E(x,y,z,t)$, $J(x,y,z,t)$, force direction, coil reaction force, joule heating, fringe losses, reactive power and structural load spectra.

G2 VALIDATION BASIS

- Plasma-off background field first; do not call it vacuum unless the chamber is evacuated
- Map B_x , B_y , B_z , magnitude and phase over the interaction volume
- Measure complex coil-impedance matrix, power factor, harmonics and thermal loss
- Include duct, cryostat, shield, seams and fasteners before combined HTS testing
- Set final limits before testing from uncertainty and design sensitivity

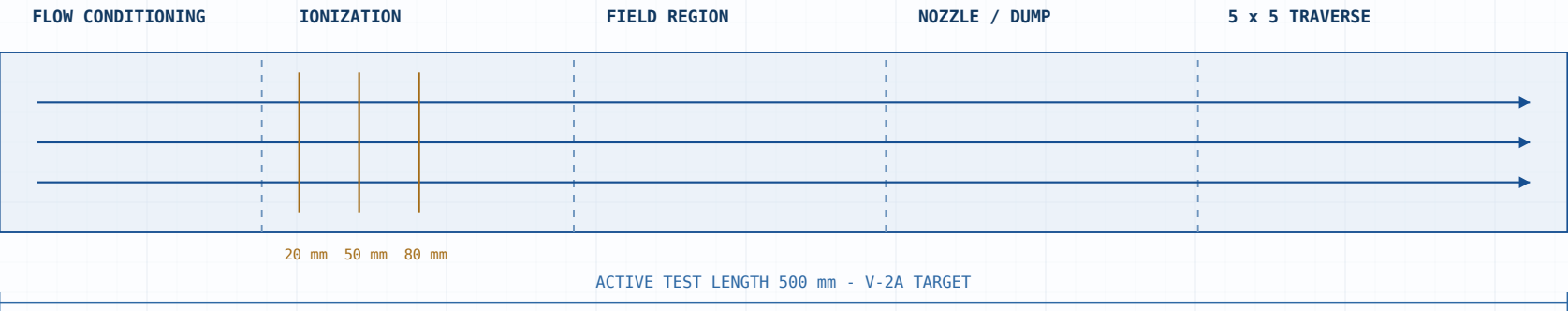
STAGE	CONFIGURATION	REQUIRED EVIDENCE	RELEASE
G2A	Numerical verification	Mesh convergence; energy conservation; small div B residual; correct materials, conductivities and circuit boundaries	Model ready for correlation
G2B	Bare segmented stator	125-point vector field grid plus seam/end points; $Z_{ij}(f)$, real/reactive power, phase and frequency sweeps	Winding candidate retained
G2C	Installed passive hardware	Repeat with duct, cryostat, supports, shield and fasteners; quantify eddy-current and shielding changes	Installation effects bounded
G2D	HTS + stator, plasma off	Combined field map, HTS temperature margin, AC loss and stored-energy behavior	Combined magnet test passed
G2E	Plasma-loaded system	Loaded impedance, fields, power and commanded force direction after conductivity gate	Electromagnetic gate closed

PROVISIONAL OBJECTIVES ONLY: FIELD RMS $\leq 5\%$; MAX LOCAL $\leq 10\%$; PHASE ≤ 5 DEG; REAL POWER $\leq 10\%$. PRE-REGISTER FINAL LIMITS.

06 MEASURE CONDUCTIVITY BEFORE SCALING

BEGIN WITH UNSEEDED NONREACTIVE AIR; ANY HOT/SEEDED BENCHMARK REMAINS A SEPARATE QUALIFIED-LAB BRANCH

V-2A INSTRUMENTED CHANNEL



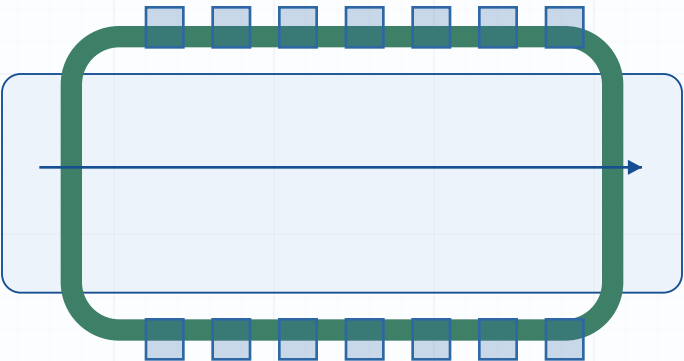
MEASUREMENT	PURPOSE	CONTROL	OUTPUT
Multi-length impedance	Separate bulk conductivity from contact/sheath effects	Plasma-off baseline	$\sigma_{eff}(x,t)$
Voltage / current / phase	Account for real ionizer and winding power	Calibrated probes	P_{in} with uncertainty
Electron / gas diagnostics	Characterize ionization and heating	Method-specific calibration	n_e, T_e, T_g
Pressure and velocity traverse	Close momentum and pressure balance	5 x 5 exit grid	$\Delta p, u(x,y)$
Decay and residence time	Determine whether conductivity survives interaction	Ionizer timing sweep	$\tau_{\sigma} / \tau_{flow}$
Force stand	Detect module-level reaction force	Blank / reversal matrix	Net force +/- uncertainty

07 HTS BIAS MAGNET ENGINEERING

THE CITED 0.2 T VACUUM THRUSTER VALIDATES A TECHNOLOGY CLASS - NOT THE V-2 FIELD-VOLUME BUDGET

CANDIDATE HYBRID MAGNET

CONDUCTION-COOLED HTS DC BIAS COIL



WARM SEGMENTED TRAVELING-WAVE STATOR

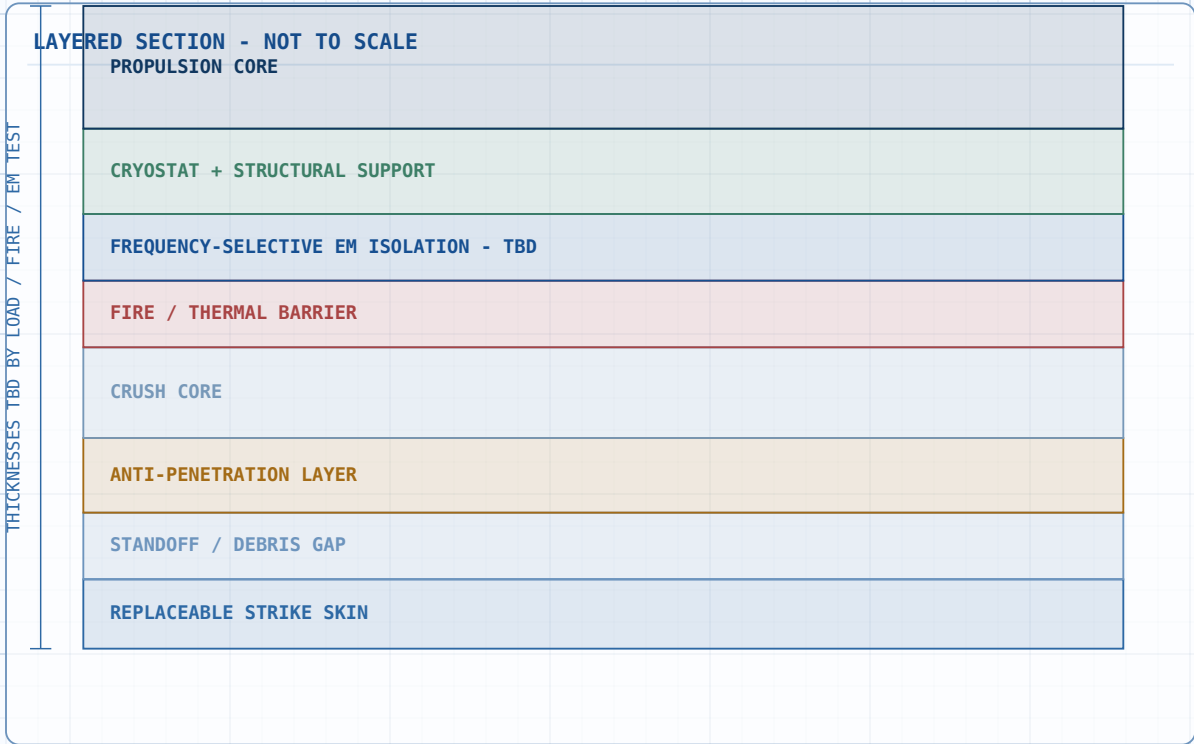
TRADE-STUDY OUTPUTS

- Required B-field map and homogeneity over actual flow volume
- HTS tape length, critical-current margin and operating temperature
- Cryocooler input, current-lead loss, radiation and conduction load
- AC loss caused by nearby traveling-wave windings
- Stored energy, quench detection, dump path and venting
- Magnetic force, pressure, fatigue, mass and service access
- External fringe-field and occupant/electronics exclusion zones

GATE	REQUIRED EVIDENCE	FAILURE MODE	DISPOSITION
Field-volume	Measured map at temperature/current	Target B only at coil center	Resize or reduce target
Thermal	Steady and transient heat budget	Cryocooler overload	Duty-cycle or architecture change
AC exposure	Loss test with traveling wave active	HTS heating / margin collapse	Increase separation or shielding
Quench	Detected and safely discharged	Local damage / overpressure	Do not energize further
Mass	Complete coil+cryo+support allocation	Vehicle budget exceeded	Shared magnet or vehicle resize

08 FIELD-COMPATIBLE PROTECTIVE SHIELD

PROTECT THE PROPULSION CORE WITHOUT CREATING A CONDUCTIVE SHORTED TURN AROUND THE TRAVELING FIELD



MANDATORY DESIGN RULES

- No continuous conductive loop may encircle the time-varying field without explicit eddy-current analysis.
- EM isolation must reduce HTS AC exposure without blocking the desired field in the fluid; slots, laminations, fasteners and seams belong in the coupled model.
- Quench pressure and thermal-runaway products vent away from occupants, controls, landing gear and lift flow.
- Structural magnet reactions bypass fragile cryogenic and ionizer components.
- Shield panels are replaceable and inspectable after any strike or overtemperature event.
- Final material, thickness and bolt schedule remain unreleased pending quantified threats and loads.

PROTECT AT ALL COSTS = FAIL-SAFE + INSPECTABLE

09 V-2A EXPERIMENTAL SEQUENCE

QUALIFIED LABORATORY WORK ONLY - NO DIY HIGH-VOLTAGE, CRYOGENIC OR SEEDED-COMBUSTION TESTING

01

INSTRUMENT

Calibrate force, flow, pressure, voltage, current, phase and temperature channels.

02

BASELINE

Run airflow with plasma and magnetic fields off; record drag, vibration and electrical noise.

03

IONIZE

Map $\sigma(x,t)$, decay, temperature and total ionizer power with magnetic fields off.

04

STATIC FIELD

Add bias field in stepped increments; check heating, force artifacts and diagnostics.

05

TRAVELING WAVE

Sweep phase, frequency, magnetic wavelength and slip at fixed flow conditions.

06

NOZZLE

Measure pressure recovery and far-wake momentum with variable outlet geometry.

07

REVERSALS

Reverse phase sequence, field polarity and commanded force; verify corresponding response.

08

CLOSE BALANCE

Reconcile field power, ionization, heat, pressure, momentum, wall loads and reaction force.

STOP ON UNCOMMANDED FORCE, INSULATION FAULT, OVERPRESSURE, OVERCURRENT, OVERHEAT, QUENCH, SENSOR DISAGREEMENT OR PERSONNEL INTERLOCK.

10 EVIDENCE GATES

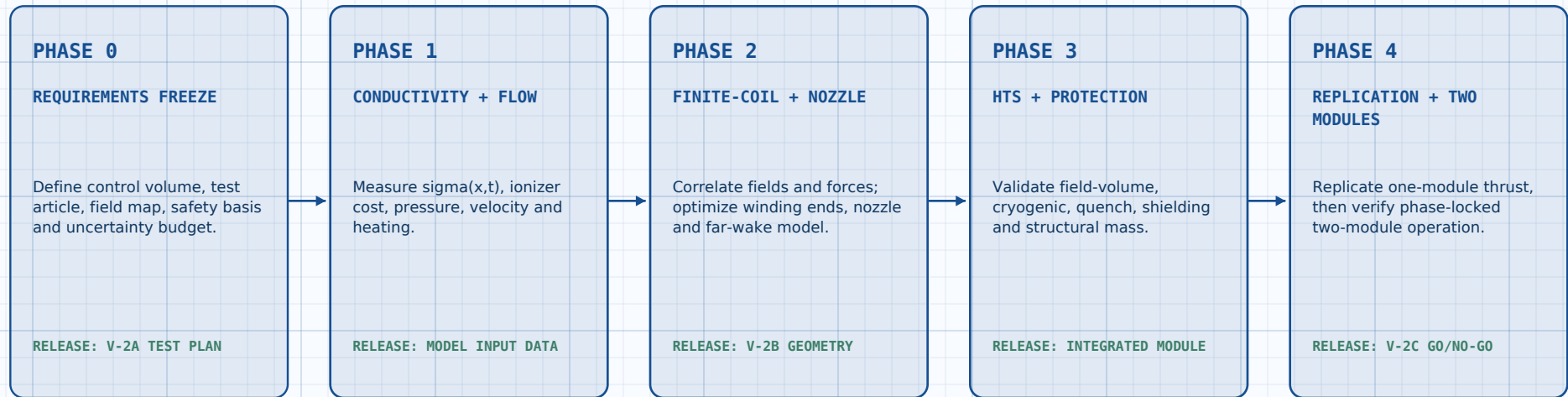
A FOUR-MODULE VEHICLE IS PROHIBITED UNTIL ONE MODULE CLOSES THE FULL EXTERNAL FORCE AND ENERGY BALANCE

GATE	PASS EVIDENCE	CURRENT CONDITION	DECISION
G1 Conductivity	Repeatable $\sigma(x,t)$ with uncertainty and total ionization power	150 S/m placeholder only	Continue / revise physics
G2 Electromagnetics	G2A-G2E: mesh-converged model correlates vector field, phase, Z matrix, power and installed losses	Scalar uniform-field model	Close staged EM gate
G3 Coupling	Force direction and magnitude follow phase/polarity controls	Not demonstrated	Reject artifacts
G4 Nozzle	Pressure rise converts to measured exit momentum	Parametric Model 0.2 only	Select geometry
G5 Net thrust	External stand force exceeds combined uncertainty and survives controls	Not demonstrated	Authorize V-2B
G6 Energy	All electrical inputs and thermal losses reconcile with fluid energy	Auxiliary/ionizer incomplete	Efficiency claim allowed
G7 HTS	Field-volume, mass, cryogenic and quench requirements passed	Candidate technology	Freeze magnet design
G8 Protection	Fire, fragment, pressure, EM and structural tests passed	Layer concept only	Release shield design
G9 Independent test	Outside laboratory reproduces the one-module result	Not performed	Authorize V-2B+
G10 Synchronization	Two modules phase-lock without destructive coupling	Not performed	Authorize V-2C

CLAIM RULE
NO 'PROPULSION DEMONSTRATED' CLAIM UNTIL G3-G6 PASS WITH TRACEABLE UNCERTAINTY AND THE COMPLETE TEST ARTICLE ON AN EXTERNAL FORCE STAND.

11 V-2 PROGRAM ROADMAP

THE GRANT ASK FUNDS DE-RISKING EXPERIMENTS - NOT A FLIGHT-READY CRAFT



CONTROLLED BASELINE STATEMENT

CRTFE V-2 is a proposed dual-field atmospheric MHD research architecture. Existing HTS and MHD research supports investigation of its component technologies, but does not validate the V-2 field volume, conductivity, thrust, power, mass, safety or vehicle feasibility.

MEASURED

No V-2 propulsion performance has yet been measured.

NEXT

Release the V-2A test plan and instrument uncertainty budget.

Primary precedents: Zheng et al., National Science Review, 2026, doi:10.1093/nsr/nwaf589.
Macheret, Shneider and Miles, cold hypersonic flow with external ionizers, AIAA conference paper, 1999.
Schulz, Chapman and Rhodes, MHD-augmented chemical rocket performance analysis, AIAA-92-3001, 1992.
Verification framework: ASME V&V / UQ standards and pre-registered measurement uncertainty.
Internal basis: CRTFE 1D Coupled Model 0.2 and CRTFE V5 preliminary engineering package.
Assurance context: FAA AC 20-115D (software), AC 20-152A (hardware), AC 20-174 (systems), AC 20-158B (HIRF).